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To cite this article: Horacio Vaggione (1993) Determinism and the false collective about models of time in early computer-aided composition, *Contemporary Music Review*, 7:2, 91-104, DOI: [10.1080/07494469300640061](https://doi.org/10.1080/07494469300640061)

To link to this article: <https://doi.org/10.1080/07494469300640061>



Published online: 21 Aug 2009.



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Determinism and the false collective about models of time in early computer-aided composition

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The late John Myhill's talk, delivered at Northwestern University's 1978 International Computer Music Conference (Myhill 1979a, 1979b), contains some interesting ideas concerning the meaning of such pairs of oppositions as stochasticity vs. determinism, periodicity vs. non-periodicity, and culture-rooted vs. "trans-cultural" facts, in relation to such problems as stylistic coherence and the nature of music.

Being a mathematician interested in musical matters as well as in philosophical questioning, Myhill was aware of the multiple levels on which music seems to be articulated. This awareness lead him to formulate questions about the musical meaning of formal procedures used by such composers as Hiller and Xenakis, contributing in this way to elevating discussion within the field of early computer-aided composition.

I believe that this conceptual universe can be revisited today, in order to clarify the role of the computer in a more highly developed compositional environment. Accordingly, the purpose of this essay is to recall some ideas Myhill exposed in his talk, accompanied by a cluster of references to other thinkers and musicians, as well as some extensions reflecting my own opinions on the subject.

1.

"Although there is probably a pretty general consensus (modulo some minor musical and mathematical details) as to what a random sequence of pitches or durations is", there remains a problem in finding "what one could possibly mean by deterministic music". (*) To put it roughly, there "seems no objective reason why we should contrast 'random pitches' with, e.g., 18th century common-practice at the other end of the spectrum, rather than with, say, 16th century polyphony, dodecaphony or the ancient Korean style".

Consequently, it must be admitted that there is "a fairly well defined notion of random pitch, but seemingly no culture-independent notion of non-random or deterministic pitch-structure".

*in what follows, words between quotation marks are direct renderings of Myhill's text, unless indicated otherwise.

So, though we have an idea of what it would be like to make music 'more and more' random (for example by increasing fragmentation) "what we mean by making music 'less and less' random seems totally culture bound and lacking anything approaching a precise mathematical definition". As an example of the difficulty of achieving the second task, Myhill refers to Lejaren Hiller's solution – see for example the *Computer Cantata* (Hiller and Baker 1964) – based in the idea that "a non-random melody consists of a mere reiteration of one frequency". This solution seems at first glance to make "impeccable musical and mathematical sense". However, Myhill adds: "I personally would feel happier with something more sophisticated and subtle than this".

Where did Myhill first turn, in order to build a model of "deterministic music" – and further on to make it coexist with the already well defined indeterministic model? To something closer to his assumption that there is no culture-independent notion of non-random structures. Thus he wondered about defining computer instructions like:

write a lute piece which begins in the manner of Dowland and 'modulates' to the style of 18th century Japanese koto music".

Consequently, he considered that "a starting point in this direction might be the analysis of Pousseur's *Wild Horse Ride* which takes us from Mozart and early Beethoven via Schubert, Brahms and Wagner to Schoenberg and Webern". But Pousseur's procedures proved to be quite "arbitrary", in the manner of culture-dependent behaviour, and appeared as intuitive and difficult to formalize.

So Myhill declared himself unable, in spite of some elementary suspicions, to find the proper way to describe them to the computer. "I know the 'modulations' are implemented by means of certain permutations of integers 1 through 12 which take, e.g., major triads after a sufficient number of applications into 'Webern-triads' like CFB or CF#B, but have no idea whether this technique can be programmed". However, Max Mathews was already in the sixties proposing a structural cross-fade between popular tunes via interpolation techniques. Mathews was at this moment explicitly concerned with an "algebra for combining functions" which he thought "specially useful for computer-aided composing" (Mathews and Rossler 1968, reprinted 1972). By means of this "simple" algebra, the computer can, for example, "average between two melodic or rhythmic lines or gradually convert one rhythmic pattern into another" (ibid.) A first example of application of this technique was the one in which *The British Grenadier* is gradually converted into *When Johnny Comes Marching Home* and back – in Mathews's terms, "a nauseating musical experience but one not without interest, particularly in the rhythmic conversions" (ibid.). This direction, which has been growing in the meantime (see Barrière 1990 for an account of the latest developments) is concerned primarily with the means to realize gradual exchanges of parametrical data using interpolation algorithms. However, it does not face the problem concerning an explicit formalization of culture-rooted musical expressions, considered as one of the poles of the spectrum between randomness and determinism.

At this point, an insidious question might arise: are we reaching here the limits of the possibilities of formalization of complexity by means of the definition of

rules? Should we enter, if this is the case (but this reveals a new paradigm being explored today), into the realm of connectionism (Loy 1989, Todd 1989) in order to speculate on the possibility of finding an answer by way of holism rather than of formal analysis?

This is indeed a very old dichotomy, in fact as much as theoretical reason itself. But the times have been bringing a plethora of new situations, especially those related to symbolic computation and cognitive science. There is an important recent polemic on this question between Otto Laske and some "connectionist composers"; yet surely we are at the very beginning of this venture (see Laske 1990). The connectionist approach looks to embrace a whole thing in a non-analytical manner (in a "brute force" strategy), prescribing to the machine (to the neuronal computer) to "learn", that is, to clone a given musical form and to reproduce it later with the desired variants. But, as Laske's polemic shows, there is at least an argument running against this procedure, an argument that stresses the lack of invention allowed by the paradigm itself. Will composers, as Laske asks, be merely copying existing musical objects and no longer creating new forms through the invention of new rules?

Since Myhill was working with the formal definition of rules, he was less than deeply concerned with the problem of cultural complexity, in spite of the fact that he was able to see it. Questioning himself about the pertinence of his computational tools to deal with the whole question* he chose rather to concentrate on "something less unapproachable, namely the contrast of regularity and randomness in regard to rhythm. Here the other end of the spectrum is simply and trans-culturally periodicity". We will see how this "something less unapproachable" indeed contained the entire problem. But as he focused his talk on this matter, I will begin my comments by referring to some basic concepts concerning music and time.

2.

"Does music exist in time or does time exist in music?" This question, as put by Jonathan Kramer in his book *The Time of Music* (Kramer 1988), is not "simply a semantic game", since "if we believe primarily that music exists in time, then we take time as an absolute, as an external reality, as somehow apart from the experiences it contains"; and on the other side, "if we believe in the time that exists uniquely in music, then we begin to glimpse the power of music to create, alter, distort, or even destroy time itself". In this latter case music can be viewed

*In fact Myhill's questioning seems to stand directly against the assumptions of cognitive science, at least in its pure computational direction. A recent paper by Peter Kugel, appropriately titled "Myhill's Thesis: There's More than Computing in Musical Thinking" (in *Computer Music Journal* 14(3), Cambridge, MIT Press, 1990) gives us some cues to clarify the core of the problem as envisaged by Myhill, as well as to understand the reasons by which "Myhill's Thesis" has been largely ignored. Unfortunately, Kugel's article appeared when my paper was already written; thus I cannot incorporate here these developments. But it would be interesting to note that his source comes from a paper presented by Myhill in 1952: "Some Philosophical Implications of Mathematical Logic: Three Classes of Ideas" (in *Review of Metaphysics* 6(2): 165-198).

as “a series of events that not only contains time but also shape it” (Kramer 1988, p. 5).

Both sides of Kramer’s dichotomy have had their adherents in aesthetics and music theory. As an example of the use of the idea of time as an active force in itself, we can recall Victor Zuckerkandl’s *Sound and Symbol: Music and the External World* (Zuckerkandl 1956). And as an early example of theorization of “musical time” as an independent force existing only in and through music, we can cite Gisèle Brelet’s *Le temps musical* (Brelet 1949).

However, there is still another possibility to be considered, manifested in the thesis (see Charles 1978, 1987) that affirms the status of co-existence (we can also say inter-penetration, or, in more logical terms, non-contradiction) of time as external force and time as shaped by musical syntax. Further support for this thesis can be found in the writings of J.T. Fraser – cited by Kramer – who holds that “time is not bound by the law of contradiction” (see for example Fraser 1975). Kramer states as much when he says that “time can be many things, and it can be them at once”. Similarly, we can recall the demonstration by I. Prigogine of the inherent multiplicity of time’s functionalities (Prigogine 1980). However, what is important for our present subject is Kramer’s assumption that “through time, music’s meanings become both internal (syntactical) and external (symbolic)” since it is “from the conflict between time taken and time evoked” that comes “the richness of musical meanings” (Kramer 1988, p. 15). I agree with this last orientation, although I would prefer not to qualify the opposition in terms of a conflict, for this would reduce them – and with them, the whole “richness of music meanings” – to the reasoning mechanism of pure dialectics. However, I believe that Kramer refers here not to the reduction of a plurality into a synthetic unity, but rather to the possibility of articulating this plurality into different interactive levels. If this is so, the more we take into account these different interactive levels, the more we preserve the “richness of music’s meanings”. A key concept appears here, on which I will insist throughout this paper: that of interaction. Assuming different and simultaneous temporalities would not have any concrete consequence – other than to show a dubious way of thinking, having fallen into “the seduction of the unarticulated” (to use Michel Bernard’s expression – but see Charles 1978) if not by the presence of the concept of interaction, which opens for us the ground to articulate – to structure, to compose – the different functionalities of time.

Time “taken”, that is, “external”, occupied, filled with something. And, on the other side, time evoked, “internal” (to musical form), created, syntactical time. In order to avoid any reductionism by the way of a dialectic synthesis, we can consider them as poles that appear in the building of the idea of music, as we look to understand it as a complex phenomenon. Once this is assumed, we can consider the “internal meaning” of musical time as belonging to music itself (as it is expressed in music), that is, belonging to the “syntactical” nature of music, which can be equated with form. Now, if we hold – as I do – the idea that, in music, form and content are indivisible, then we can consider all temporalities present in music in a non-contradictory, pluralistic manner, and still retain syntactical time in order to articulate them all – because any articulation (any articulated thinking) is expressed in form, and form is precisely the result of a multi-level interactive articulation. I will stress in a moment the difficulty of doing this within a

paradigm (information theory) that refers to only one level of articulation, and that is therefore unable to integrate "the richness of music's meanings". In this way I intend to approach, avoiding reductionism, the polarities of musical time referred to by Myhill.

Hence, our central question will be: how to understand Myhill's problem (namely, a lack of symmetry between a well-defined working concept of indeterminacy and a fuzzy culture-dependent idea of determinism) in relation to our concern with syntactical time? And, furthermore: how is syntactical time actually shaped by means of both stochastic and deterministic strategies?

3.

But perhaps it will be pertinent at this point to recall Wittgenstein's famous remarks: there is always a "form of life" involved in the emergence of any "language game"; thus, any syntactical universe is by nature an expression of "the culture of one period". This is the major concern of the *Philosophical Investigations* (Wittgenstein 1953), which cannot be regarded as an expression of crude historicism, or crude organicism, causalism, or any of the orientations traditionally related to determinism, but rather as a formulation of a non-metaphysical primitive, stating a kind of "neither arbitrary nor grounded" status in respect to the nature of any formalized thinking.

Now, if we create an example in the manner of the "imaginary anthropology" which Wittgenstein delighted so much in using, and imagine any musician of the past centuries coming to our time and listening to some electronic music piece and asking for its function and meaning (that is, for an explanation in terms of his own universe): our most straightforward answer to this situation would be "we call this *music*" (but see Bouveresse 1971, 1988, for a discussion of this "ungrounded reality" in relation to the status of creativity and cultural complexity).

In any case, the pertinent question (for our subject) would be: given the complexity of cultural practices (and taking into account the fact that formal tools – of any kind – are part of these practices), can we build up a set of rules to approach this complexity? If we answer in the negative, then we must state the existence of an intangible borderline between "trans-cultural" (infra-cultural?) formalism and cultural multi-level complexity. Or, if we choose the contrary option, then we must ask: is there a continuum between these two poles, and, if it is the case, should we re-present it? Or should we rather talk uniquely about the appropriate "sampling rate" to be used to match the complexity of anything?

This is the not-so-simple problem faced by Myhill when he asked about the other pole opposed to indeterminacy, and concluded that there is no clear formal definition of what a deterministic sequence might be: "there seems no objective reason" why we should contrast "random pitches" with a certain culture-rooted musical system rather than with any other. But precisely this is what, after Wittgenstein, stands at the heart of culture-rooted phenomena: they are neither arbitrary nor grounded in some kind of objective causal reason; and they reveal, by this very fact, that they belong to a particular "style of thinking" (Wittgenstein 1966).

4.

The problem faced by Hiller while composing the *Computer Cantata* by means of stochastic algorithms was: what might be the opposite pole to a totally random melody? Or, in other terms: what might be a totally deterministic melody? The solution he found was to reduce the variety of pitches as they appear in a random distribution to the repeated enunciation of only one pitch. Doing so, Hiller equated total randomness with maximal information, and, conversely, total redundancy with pure determinism, following the paradigm of information theory (Hiller and Isaacson 1959, Hiller and Fuller 1967). Of course, a totally random melody contains the minimum of predictability: hence, if we keep reasoning within this paradigm, the maximum of predictability must be equated with zero information. In this way, information theory equates the pluralistic appearance of the living universe with diverse orders of probability. Therefore a probability of order zero would maximize information because all terms would be weighted with equal probability. And the existence of very singular things, such as the musical systems in the world's music, are manifestations of particular weighted orders of probability between zero and one.

The first thing to notice in this description is the crossing of the two scales of predictability and redundancy. Thus, zero-predictability (zero-order in a Markov chain) equals maximum of information, and hence absence of regularities or periodicities, e.g., chaos. The opposite pole of chaos would be total order, equated with total redundancy, thus with zero-information. Life, and all existing things, would be somewhere in the middle, theoretically described as particular weighted probabilities of first, second, third . . . nth order Markov chains. Reasoning along these lines, Hiller equated a totally deterministic melody with the repetition of only one pitch, following the equation *zero information = total predictability*, that is, *total redundancy = total determinism*.

I have already alluded to the Myhill's claims for "something more sophisticated and subtle than this". So what is wrong with Hiller's information-theoretic solution? There are here several answers. To begin with, as Kramer points out, "it is impossible in practice to specify a maximum order (in Markov chains) that would account for all meaningful probabilities in a given composition" (Kramer 1988, citing Wayne Slawson). And, more specifically, "a major problem of information-theoretic analysis of music is the impossibility of isolating the appropriate Markov order. If event C depends on even B and B depends on A, then there may nonetheless be a functional relationship between A and C, despite an apparent first-order construction" (Kramer 1988, citing Justin London). For instance, we can easily consider the case in which a sequence consisting of one repeated pitch can be non-predictable within the musical context in which it appears. That is, the orders of probability can be not only musically irrelevant, but logically incoherent, and hence totally dissociated from the arrangement to which they seem theoretically to belong.

I believe that we are facing here a conflict of dimensions, or, as I have said before, of levels of articulation. Information theory has failed totally in its attempts to explicate any structure of real complexity, because it allows the description of only one level of articulation. And there seems to be no possibility of dealing simultaneously with more than one level of articulation because the

paradigm inside of which the theory works is concerned only with an a-syntactical (or infra-syntactical) approach to the facts that it describes. However, the limitations of information theory to deal with complexity – and especially with culture-rooted complexity – seem to be a case of a more general situation, concerning epistemology in its broader sense. A further demonstration can be taken from fractal processes (Mandelbrot 1975). These processes generate, through the iteration of a simple formula (a fractal code), a form exhibiting an infinity of details: but these details are not created as singularities but produced by a global causal law.

I am not denying the utility of these procedures, in that they are used by scientists looking for regularities. A composer, however, knows how to generate singular events, and how to articulate them in bigger and bigger chunks without losing the control of the singularities: this is what Myhill calls stylistic coherence, and Kramer syntax. This is why it is always problematic to utilize global causal laws (e.g., stochastics) in music composition, if their automatism is not compensated by compositional choices concerning other levels of articulation. This is why Xenakis, after being concerned with Markov chains, subsequently adopted a consistent silence about his more recent compositional procedures, a silence not broken by his claim that he has introduced an “arbitrary” (that is “composed”) manipulation of the data provided by his stochastic canons (Xenakis 1979).

5.

Curiously, Myhill comes (in his talk) to consider the psychological aspects of the duality determinism/indeterminism through a straightforward reference to Hans Eisler’s writings about the “false collective”. In what follows, I will quote directly the rendering of Eisler’s ideas in Myhill’s words:

Eisler states as a problem (in his book about film music): how does one make music to describe the mobilization of an army? At the beginning one sees, say, a mobilization notice and observes people reading it, going home, preparing to depart for the war, bidding farewell to their families, etc. The subjectivity is shocked, and erratically numbed or painful. As they go out into the streets, their subjectivity becomes more organized and anonymous, they think of themselves less and less as individuals, more and more as parts of One Thing (the *Böse Kollektivität*)”.

However, I am not echoing here Eisler’s problem in order to raise any extra-musical (meta-symbolic) agenda. It is with the formal (syntactical) implications of this situation that I intend to deal. Myhill himself cites this problem in the context of his efforts to deal with stochasticity versus determinism in the reduced field of rhythmic relationships, as a preliminary step in the search for a “continuum” between these two poles. I will refer to these efforts in a moment. But first let me recall the example of Xenakis – see his book *Formalized Music* (Xenakis 1963) – which makes a parallel between stochastic laws and the kind of global phenomena found in nature, such as the noise of rain, or that of bubbles of champagne, or street noises, or movements of a human mass in a street political demonstration, etc. – which he claims are “described” and “controlled” by Poisson’s statistical law of distribution.

Xenakis states that this kind of global law gave him the inspiration to bypass the "crisis of serial music" (which he explicates by the existence, in totally-ordered serial works, of an excess of complexity that destroys the sense of linear polyphony) by proposing the concept of mass-polyphony, which consists roughly of the application to musical processes of such formal descriptions of natural global phenomena. But in doing so, Xenakis eliminated theoretically the possibility of a truly polyphonic, multilayered discourse, restraining himself to a kind of homophony (his "clouds" or uniform textures). Manipulating sound events as statistical "populations", Xenakis negates the singularity of these events, retaining only their external, one-sided aspect of elements governed by a global law, and hence devoid of any intrinsic formal property.

What is also important, however, is the fact that Xenakis tried in this manner to compose a "trans-cultural" music, in the sense stated by Myhill. Xenakis himself cites the reaction of Hermann Scherchen concerning *Metastasis*, that ran more or less like this: what is interesting in this work is that it seems to come from another world different from the world of music (Xenakis 1979). That is, this work is interesting ("new") because it comes not from the cultural but from the natural (trans-cultural) side of reality, of which it seems to be more than a metaphor, but a kind of direct translation.

But despite this approach being effectively new for its time, it was to be surpassed by the composer himself. So from the "polyphony of masses" Xenakis moved on to integrate into his compositional world the other pole, the "arbitrary", that is, the musically composed, as opposed to the stochastically generated. I used to refer to this other pole, by way of illustration, equating it with the so-called "cocktail-party effect", that, as everybody knows, describes a situation (roughly opposed to global stochasticity) in which is involved an act of direct, focalized perception, that is, an act of selection of singularities out of a given statistical globality.

At this point, I remember a conversation at the University of Paris between Messiaen and Xenakis: the former explained to the latter that, while collecting "bird songs" in the forest, he used simultaneously two media: music written on paper and the tape recorder. Once at home, he was always struck by the difference of content between the two documents, the recorded tape being a kind of global soundscape, while the written examples corresponded to the composer's reading of his own focalized perception, concentrated on some special events of this soundscape. Messiaen was always surprised to hear the recorded soundscape, because he did not remember having heard it at all when he was in the forest; he had only selected the bird calls that had attracted his attention.

6.

So our question now is: can we equate the oriented, selective listening we have referred to with determinism? Can we say that it is a deterministic listening? Determinism would have here the meaning of: to focalize a specific - singular - gestalt.

Messiaen's way of listening is indeed not very different from his way of improvising at the organ or writing his compositions. We are always confronted

with an activity of selection. To formalize – and Messiaen had been formalizing quite a lot in his compositional microworld – is to select rules of selection, in other words, to make choices and to make rules that allow certain choices to become real.

To compose would mean therefore to exercise a generalized activity of selection, comprising all possible levels of articulation and all scales of time (from the micro-local to the macro-global). But this activity is only generalizable if what is to be selected corresponds to singular events (of any size), and if the rules chosen to make selections at any level of a time scale are interactively connected and implicated. Any formalization that regards only a single level, for example a single non-interactive law of global distribution of points into lineal time, is simplistic and reveals a flat way of thinking.

The statement above, formulated in a very abrupt fashion, certainly does not correspond to the manner in which Xenakis considered the compositional process. In this context, there is one point in my critique of Xenakis' early ideas that needs to be stressed: Xenakis negates the status of singularities of the events he manipulates, considered only as "populations" submitted to a global law. By retaining only the external aspect, namely, the capacity to be governed by a global law, he rendered them devoid of any intrinsic formal properties. But this is only one side of his composing strategy. The other side, which appears in his works more and more clearly after *Eonta* is, as I have said, the "arbitrary" activity of (deterministic?) selection.

In fact, what Xenakis has been doing can be related to the problem stated by Myhill in his talk: a quest for a continuum between stochasticity and determinism. Xenakis, in his "Opening Address" to the same 1978 International Computer Music Conference at Northwestern University, called it "the new diapaisson". This was the object of Myhill's reflections: "in this continuum, randomness or stochasticity is one end, and there is no such uniformity, given the density of musical cultures and styles, as to what to put at the other end". But in spite of this awareness, Myhill's considerations centered "on the definition of a music that would be at once deterministic and stochastic, or else stochastic in some parts and deterministic in other parts, without any loss of stylistic coherence".

The two last statements are indeed contradictory, because if we do not know what a deterministic music might be, how would it be possible for us to affirm the said continuum?

And, if we admit that the so-called deterministic side of the continuum should be equated with arbitrary selection, then are we not cutting the compositional process between a one-sided formalization and an unrelated and inconsequential procedure of selection? If we negate a singular event's capacity to contain intrinsic formal properties, from whence might the "stylistic coherence" come?

7.

Myhill claimed to have been assisted in his talk by "a serendipitous cross-fertilization between Xenakis and Eisler". So, after referring to Eisler's problem as stated in the "False Collective", Myhill comments: "I ask myself, specializing Eisler's questions, how this is reflected in the rhythmical character of the

accompanying music" (of a film scene dealing with the transition between the mobilization of the individual and its integration in the *Böse Kollektivität*). The answer runs: "increase the degree of periodicity!"

So, if we understand this example, the continuum thus envisioned runs from the aperiodic rhythms manifested by an individual behaviour (globally viewed as a disorder, but coherent at the level of each singularity) to the periodic rhythm manifested in the behaviour of an army, appearing as One Thing, namely, the – "False" – Collective, globally viewed as an order where singularities are negated as to give coherence to the whole. Of course, Eisler can qualify this whole as a "false" collective, because it is one-sided: a whole formed not by the qualities of its parts, but standing uniquely by its self-affirmation as a whole.

Is this false collective a good model of determinism? Can the increase of periodicity – which I agree well represents an increase of homogeneity, and subsequently a decrease of diversity – be said to be (more) deterministic? Only in that it is (more) simple. But in equating determinism with simplicity we remain inside Hiller's model that assimilates pure determinism with pure redundancy. And thus we are coming back to the "not-so-subtle" solution that Myhill was trying to avoid.

At this point, I would like to ask: which is the model of time of the marching of an army? For the sergeant charged with the order of marching, the scheme is "one-two", as it is also for the soldier who must follow this order and hence count "one" and "two" to be integrated in what Zuckerkandl calls "a metric wave". To understand this point we can recall Zuckerkandl's thesis about music and time. As explained by Charles: "if in the simplest of metric waves, namely the two-phase wave, 'two' (present) follows 'one' (past), the 'past' character of 'one' is not synonymous with non-existence (even if, according to the hourglass concept of time, such a 'past' ought to be already extinguished); if 'one-as-past' were extinguished, 'two' would simply be a second 'one'. The wave says 'one-and-two', not 'one-or-two'. The past is never absent from the wave-as-a-whole. And on the other hand, the 'future' two is no more absent when one beats the 'one' of the beginning of the wave" (Charles 1986).

This demonstration accounts for Zuckerkandl's thesis of active time, of time as a force existing in the external world, and of time as a category of reality that does not know any "equality of times". Applied to the "deterministic" music of the past, it explains why there is so much repetition (e.g., in Beethoven's *Eroica Symphony*): in music (unlike in language) things can be stated again and again, because time is "always new" (Zuckerkandl 1956; see also Vaggione 1980).

In Charles' explanation, the "simplest wave" is taken as a means to demonstrate the motion of time itself. But in order to be applied to the phenomenon of music, we must complexify it to the point of modeling a *polyphony of waves*. This polyphony makes music not a pure periodic phenomenon, but a *quasi-periodic* one. The difference is essential, as we will see in a moment, in order to characterize music as a highly interactive field of co-present temporalities.

In the case of the marching army, however, we remain roughly inside the simplest metric wave: one-and-two. A huge mass of people behave in a way so simple that it becomes a single sustained wave (and this is not only a metaphor: a marching army actually creates a single low frequency which has caused the collapse (by an acoustical phenomenon of "bad sympathy") of houses and bridges!).

I agree that this simple behaviour can be called deterministic because it is extremely easy to predict. However, it is better called globally ordered. The terms – the elements – involved in this globality are not taken as individuals, as singularities: their intrinsic (formal, i.e., relational) properties – the personalities of the soldiers, in this case – do not count for the whole. Only count the effectiveness of the (global) rules – as few in number as possible, and as redundant as possible, to create a total order: *only* one-and-two.

But the same can be said of a Xenakis's polyphony of masses (the concept is in itself a contradiction: a mass is by nature the contrary of a coexistence of very differentiated things). The famous "thesis of the minimum of constraints" (Xenakis 1963), which was implemented as an algorithm for his work *Achorripsis*, was certainly stochastic (and theoretically very informative, as it was dealing with equal probability). Furthermore, this stochasticity was expressed, among other things, by the aperiodicity of its rhythmic content. However, the relationship between the whole and the elements of the same nature as in the example of the marching army: no singularities – and hence no specific formal (relational) properties – are here allowed.

8.

The model of time of a marching army, as well as that of a stochastic cloud, can be regarded as a kind of global time. Concerning the marching army, the global time clearly coincides with the local time at any point, the "one-and-two" being present locally, but established globally. Global time implicates a self-similarity of the local with the global, established as a unique top-down order. But we cannot say that this order is hierarchic, because its one-to-one simplicity gives no room for any interaction to take place; so it is roughly a one-sided domination of the local by the global.

But also a stochastic cloud is imbedded in global time. In one case as in the other, an interaction of levels cannot take place. In Xenakis's manner of composing, global law generates the material; later on, this material is arbitrarily arranged in sections, providing lineal cues in order to make the clouds directional, by increasing, decreasing, or changing abruptly the values of some parameters, such as intensity, for instance. So we have here a strategy involving the definition of the global law as well as the arbitrary manipulation of the products of the global law, without however establishing an explicit structural relationship between the two levels.

By contrast, consider the case of hierarchic time relationships as they appear often in tonal music – one of the common musical practices confused with determinism. As Kramer states, "most tonal music is metrically regular on the surface level. On deeper levels, however, irregularities are common. Relatively rare is the piece that consists entirely of four-bar phrases grouped into eight-bar phrase pairs, 16-bar periods, and 32-bar sections". In other words, the (local) time of a phrase is not often "exactly coextensive with a four-bar 'hypermeasure'". This formulation leads Kramer to state that, in the most general case, "a phrase is not a large-scale measure but rather a rhythmic group" (Kramer 1988, p. 83). Hence, a phrase is not simply self-similar to the whole or to any hypermeasure (global time): the hierarchy implicated by the ensemble of levels of articulation is

one on which these levels, being not simply self-similar, are clearly interactive. A further demonstration comes if we consider that, while "a measure is cyclic, in that after the music has moved through beats 1,2,3, and 4 (for example), it goes back to (another) beat 1", the "rhythmic groups are not usually cyclic, because they vary considerably and because they are comprised of music, not just beats. It is because meter is cyclic that it is more resistant to change than rhythm. Rhythm is the force of motion, while meter is the resistance to that force" (ibid.) What is alluded to here is a state of interaction between the temporalities on different levels of organization, interaction that allows the definition of formal possibilities between the two poles of the local and the global, and hence an activity of deduction of articulated configurations.

This reference to tonal metric organization is not intended to be more than it is: a remainder of the multi-level organization of a common musical practice. But it lead us to an important conclusion: in the extreme opposed to this organization we find the entire stochastic field and its inherent one-level status. Any of the "non-stochastic" (we cannot consider as pertinent the use of the term "deterministic") music practices, as the ones considered by Myhill, is organized in a plurality of levels which contribute interactively to the emergence of a highly articulated form.

Confronted with the impossibility of establishing a continuum between randomness and determinism, Myhill's questions actually show that music is not a one-dimensional fact.

It is in this perspective that the efforts of the pioneers of computer sound synthesis (Mathews 1969, Risset 1969) are to be placed. Their efforts were very important to the emergence of compositional microworlds, understood as generally concerned with the shaping of time on all possible levels, including the temporal structure of the sound itself (for an account covering the varieties of approaches dealing with micro-time, see Risset 1986). Mathews's "logic for combining functions", mentioned early in this paper, was conceived as an extension of the sound synthesis program MUSIC V, pointing to the use of the digital computer as a general tool in every domain concerning music structuring. And it is in regard to the entire compositional process that the shaping of the micro-time of sound is to be understood – integrated in this process as a carrier of form.

9.

In composing music, we are composing time, that is, we are articulating it into a variety of temporalities. I believe that this affirmation acquires a very concrete meaning through the use of the computer, which constitutes a tool for shaping time at all possible levels.

I have been concerned here with the problem of computer-aided composition at the beginning of its history. I have directed the attention of the reader – using as a frame Myhill's comments – to some points that were not much discussed in those early times. I think that this discussion is important in order to clarify the role of the computer in a more developed compositional environment. The relation between determinism and stochasticity is a very important part of this clarification,

as it is the nature and the role of computational rules in the understanding and the realization of generative processes. If music is to be increasingly formalized (see Barrière 1990), we need to know where this formalization can take place, and, fundamentally, what this formalization means. Simple automated techniques of event generation, simple symbolic manipulation by means of linear computing mechanisms, are not concerned enough with the phenomenon of music; this is the reason why I stressed the "not-so-simple" aspects of formalization.

I have emphasized the temporal aspects of this problem, looking forward to the ultimate use of the computer as a tool for articulating time. As this argument has taken many pages, I will not refer here to the current work in the field (but see Vaggione 1984, 1987, 1989). I would rather like to conclude by an affirmation: the role of the computer in the shaping of musical time – which must be viewed as being the matter of computer-aided composition – needs to be further worked out, from the temporal behaviour of the partials of a spectrum, to the multiple temporalities involved in a complex sound-object, to the macro-time of the (causal and/or emergent, lineal and/or non-lineal) global form, without forgetting all the fractional dimensions in between, nor the structural relationships established at each level by interaction with all other levels.

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